

Requirement Analysis Report — SCRUM-101

1. Document Information

Field	Value
Requirement ID	SCRUM-101
Title	E-commerce Checkout Process
Document Type	User Story
Source File	user_stories/SCRUM-101-ecommerce-checkout.md
Application Under Test	https://www.saucedemo.com
Test Credentials	standard_user / secret_sauce
Analysis Date	2026-09-15
Analyzed By	Requirement Analysis Agent
Report Version	2.0 (re-run)

2. Business Objective

Enable a logged-in customer to complete a purchase on the SauceDemo store through a complete checkout flow: review the cart, enter shipping/checkout information, select a payment method, review the order (payment and shipping summary, subtotal/tax/total), and confirm the order. The process must be intuitive, secure, and provide clear feedback at every step.

3. Requirement Summary

The story defines a 4-stage checkout flow (Cart Review → Checkout Information Entry → Order Overview → Order Completion) plus a cross-cutting error-handling requirement, on the SauceDemo demo storefront. It is expressed as **5 acceptance criteria** (AC1–AC5) and **5 business rules** (BR1–BR5), with supporting technical notes on automation scope (Playwright, 3 browsers, mobile responsiveness) and a Definition of Done.

- Acceptance Criteria: 5
- Business Rules: 5
- Functional Requirements extracted: 15 (11 from acceptance criteria, 4 derived from business rules with no corresponding acceptance criterion)
- Non-Functional Requirements extracted: 4
- Open Questions raised: 10

4. Functional Requirements

ID	Description	Source
FR-01	The cart page must display all items added to the cart, including name, description, price, and quantity for each item.	AC1
FR-02	The cart page must display the total price calculation for the items in the cart.	AC1

ID	Description	Source
FR-03	The cart page must provide an option to continue shopping and an option to proceed to checkout.	AC1
FR-04	Clicking the "Checkout" button on the cart page must redirect the user to the checkout information page.	AC2
FR-05	The checkout information page must present form fields for First Name, Last Name, and Zip/Postal Code, all of which are mandatory.	AC2, BR1
FR-06	If any mandatory field on the checkout information page is left empty and the user clicks Continue, an error message indicating which field is required must be displayed.	AC2
FR-07	After entering valid checkout information and clicking "Continue", the user must be redirected to the checkout overview page.	AC3
FR-08	The checkout overview page must display a summary of all items in the order, payment and shipping information, and the subtotal, tax, and total amount.	AC3
FR-09	The checkout overview page must provide options to Cancel or Finish the order.	AC3
FR-10	Clicking "Finish" on the checkout overview page must redirect the user to an order confirmation page showing a success message and a "Back Home" button that returns the user to the products page.	AC4
FR-11	Entering invalid data (e.g., special characters, incomplete information) on the checkout information page must produce appropriate validation error messages and prevent the user from proceeding until all fields are valid.	AC5
FR-12	The system must restrict access to the checkout flow to authenticated (logged-in) users only.	BR2
FR-13	The system must prevent a user from proceeding to checkout when the cart is empty.	BR3
FR-14	Completing an order (order confirmation) must clear the cart.	BR4
FR-15	The system must allow a user to cancel the checkout process at any step and return to the cart.	BR5

Note: FR-12 through FR-15 are derived from Business Rules that have no corresponding acceptance criterion describing the expected UI/system behavior (see Open Questions #3 and #4). They are listed here as documented obligations, not as inventions.

5. Non-Functional Requirements

ID	Category	Description	Source	Notes
NFR-01	Compatibility	The checkout flow must be tested across Chrome, Firefox, and Safari browsers.	Technical Notes	No specific browser versions specified.
NFR-02	Responsiveness	The checkout flow must be mobile responsive.	Technical Notes	No specific device list, breakpoints, or viewport sizes specified.
NFR-03	Usability	The checkout process should be intuitive, with clear feedback provided at each step.	Story Description	Not measurable as stated; no concrete usability acceptance criterion or metric is provided. Flagged as ambiguous.

ID	Category	Description	Source	Notes
NFR-04	Security	The checkout process should be secure.	Story Description	No concrete security requirement, data-handling rule, or acceptance criterion is defined anywhere in the document. Flagged as ambiguous/gap.

6. Actors

Role	Identifier	Description
Customer / Logged-in User	standard_user	The only actor referenced in the document. Must be authenticated before checkout can be accessed (BR2). No other roles (e.g., guest user, locked-out user, problem user) are described in this story.

7. Acceptance Criteria

AC1: Cart Review

- GIVEN I am a logged-in user with items in my cart
- WHEN I navigate to the cart page
- THEN I should see all added items with their details (name, description, price, quantity)
- AND I should see the total price calculation
- AND I should have options to continue shopping or proceed to checkout

AC2: Checkout Information Entry

- GIVEN I am on the cart page with items
- WHEN I click the "Checkout" button
- THEN I should be redirected to the checkout information page
- AND I should see form fields for First Name, Last Name, and Zip/Postal Code
- AND all fields should be mandatory
- WHEN I leave any field empty and click Continue
- THEN I should see an error message indicating which field is required

AC3: Order Overview

- GIVEN I have entered valid checkout information
- WHEN I click the "Continue" button
- THEN I should be redirected to the checkout overview page
- AND I should see a summary of all items in my order
- AND I should see payment and shipping information
- AND I should see the subtotal, tax, and total amount
- AND I should have options to Cancel or Finish the order

AC4: Order Completion

- GIVEN I am on the checkout overview page
- WHEN I click the "Finish" button
- THEN I should be redirected to the order confirmation page
- AND I should see a success message confirming my order
- AND I should see a "Back Home" button to return to the products page

AC5: Error Handling

- GIVEN I am on the checkout information page
- WHEN I enter invalid data (e.g., special characters, incomplete information)
- THEN I should see appropriate validation error messages
- AND I should not be able to proceed until all fields are valid

8. Business Rules

ID	Rule (verbatim)
BR1	All checkout form fields are mandatory
BR2	Users must be logged in to access checkout
BR3	Cart cannot be empty when proceeding to checkout
BR4	Order confirmation should clear the cart
BR5	Users can cancel checkout at any step and return to cart

9. Dependencies

ID	Dependency	Notes
DEP-01	Authentication / Login feature	BR2 requires the user to be logged in before checkout is accessible; login itself is not specified in this document.
DEP-02	Cart / Product catalog feature (add-to-cart)	AC1 assumes items are already present in the cart; how items are added is out of scope of this story.
DEP-03	Payment method selection/entry	The Story Description states the user will "select payment method," but no field, page, or acceptance criterion in this document describes a payment selection UI.
DEP-04	Cross-browser and mobile test environment	Technical Notes require Chrome, Firefox, Safari, and mobile-responsive coverage.
DEP-05	SauceDemo test environment availability and stability	All acceptance criteria assume <code>https://www.saucedemo.com</code> and the <code>standard_user</code> / <code>secret_sauce</code> credentials remain valid and reachable.

10. Assumptions

ID	Assumption
ASM-01	The <code>standard_user</code> / <code>secret_sauce</code> test account is valid and remains usable in the target test environment.
ASM-02	The cart already contains at least one item before AC1 begins; the mechanism for adding items is assumed to work correctly and is not covered by this story.
ASM-03	Tax and subtotal calculation logic is pre-existing application behavior and is out of scope for validation beyond confirming the values are displayed (AC3).
ASM-04	"Payment method" and "shipping information" shown on the overview page (AC3) are assumed to be static/mock data rather than user-entered/selectable values, since no such input fields are defined anywhere in the document.

11. Risks

ID	Risk	Impact
RISK-01	NFR-04 (Security) has no concrete, testable definition.	Test cases cannot verify "secure" checkout in any meaningful way; false sense of security coverage.
RISK-02	NFR-03 (Usability/"intuitive") has no measurable criterion.	Usability assessment will be subjective and inconsistent between testers/automation.
RISK-03	Story Description references "select payment method," but no AC, field, or business rule defines this behavior.	Functional gap risk: automation built strictly from the ACs will not exercise payment-method selection, while stakeholders reading the story description may expect it.
RISK-04	No specified copy/text for validation error messages (AC2, AC5).	Automated assertions on error text may be brittle, inconsistent, or need to fall back to weaker presence-only checks.
RISK-05	BR3 (cart cannot be empty) and BR5 (cancel at any step) have no corresponding acceptance criteria describing exact system behavior.	Ambiguous expected behavior increases risk of incorrect assumptions being built into automated tests.
RISK-06	NFR-01/NFR-02 declare cross-browser and mobile scope without versions, devices, or breakpoints.	Coverage may be inconsistent or disputed as "incomplete" without an agreed test matrix.

12. Edge Cases

The following are not explicitly covered by any acceptance criterion but are implied by the business rules, technical notes, or general checkout-flow behavior, and should be flagged for QA coverage discussion rather than assumed:

- Attempting to proceed to checkout with an empty cart (BR3) — no AC defines the expected message/behavior.
- Leaving all three checkout-information fields empty simultaneously — which field's error is shown, or are multiple shown?
- Entering excessively long values into First Name, Last Name, or Zip/Postal Code (no length limits documented).
- Entering special characters, script-like strings, or emoji into the checkout-information fields (AC5 references "special characters" only generically).
- Non-numeric or international Zip/Postal Code formats (no validation rule documented).
- Browser back-button navigation during the checkout flow (Technical Notes call this out, but no AC defines expected behavior).
- Session expiration or forced logout mid-checkout (BR2 requires login, but expected behavior if the session ends mid-flow is undefined).
- Page refresh mid-checkout — whether entered data or cart state persists is undefined.
- Availability of a Cancel option on the checkout-information page specifically (AC2 does not mention one; only AC3 shows Cancel on the overview page), relevant to BR5's "cancel at any step" claim.
- Rapid/double-clicking the "Finish" button (possible duplicate order submission) — no defined idempotency behavior.
- Explicit verification that the cart is empty after order confirmation (BR4) — no AC directly checks this post-condition.

13. Open Questions

1. The Story Description says the customer will "select payment method," but no acceptance criterion, field, or business rule describes a payment-method selection UI. Is this a documentation gap, or does it describe functionality intentionally left out of scope for this story?
2. What is the exact expected error message text for each specific empty or invalid field? The document only says "an error message indicating which field is required" (AC2) or "appropriate validation error messages" (AC5), without specifying wording.

3. BR5 states users can cancel checkout "at any step," but AC2 (checkout information page) does not mention a Cancel option — only AC3 (overview page) does. Does a Cancel option exist on every step, and if so, what does it do on each?
4. BR3 states the cart cannot be empty when proceeding to checkout, but no acceptance criterion describes what happens when a user attempts this (blocked navigation? error message? redirect?).
5. What validation rules apply to the Zip/Postal Code field (numeric-only, length limits, country-specific formats)?
6. What specifically defines "secure" for this checkout flow (NFR-04)? No requirement addresses HTTPS enforcement, input sanitization, session handling, or data protection.
7. What specifically defines "intuitive" for this checkout flow (NFR-03), and what metric or criterion would be used to judge it as met?
8. Are additional user roles (e.g., SauceDemo's locked-out or problem user accounts) in scope for this story, or is `standard_user` the only actor to be tested?
9. Which specific browser versions and mobile device/viewport sizes are in scope for NFR-01 and NFR-02?
10. Does "confirm their order" (Story Title) imply any backend or order-persistence requirement beyond displaying the confirmation message and clearing the cart (AC4, BR4), or is the UI confirmation the entire scope?

14. Requirement Traceability

Item ID	Type	Source Section
FR-01, FR-02, FR-03	Functional Requirement	AC1: Cart Review
FR-04, FR-05, FR-06	Functional Requirement	AC2: Checkout Information Entry
FR-07, FR-08, FR-09	Functional Requirement	AC3: Order Overview
FR-10	Functional Requirement	AC4: Order Completion
FR-11	Functional Requirement	AC5: Error Handling
FR-12	Functional Requirement	Business Rules — BR2
FR-13	Functional Requirement	Business Rules — BR3
FR-14	Functional Requirement	Business Rules — BR4
FR-15	Functional Requirement	Business Rules — BR5
NFR-01, NFR-02	Non-Functional Requirement	Technical Notes
NFR-03, NFR-04	Non-Functional Requirement	Story Description
BR1–BR5	Business Rule	Business Rules section

Item ID	Type	Source Section
AC1–AC5	Acceptance Criteria	Acceptance Criteria section
Actor: Customer / Logged-in User	Actor	Test Credentials, BR2, AC1
DEP-01	Dependency	Inferred from BR2
DEP-02	Dependency	Inferred from AC1
DEP-03	Dependency	Inferred from Story Description vs. AC3
DEP-04	Dependency	Technical Notes
DEP-05	Dependency	Application URL, Test Credentials
RISK-01–RISK-06	Risk	Derived from NFR-03, NFR-04, Story Description, BR3, BR5, Technical Notes
Open Questions #1–#10	Gap	Derived from cross-referencing Story Description, Acceptance Criteria, Business Rules, and Technical Notes

15. QA Coverage Recommendations

- **Happy path:** Full end-to-end checkout flow from cart review through order confirmation (AC1 → AC2 → AC3 → AC4), verifying redirects, displayed data, and the success message.
- **Negative / validation testing:** Each mandatory field (First Name, Last Name, Zip/Postal Code) left empty individually and in combination (AC2); invalid data such as special characters or incomplete information (AC5).
- **Business-rule coverage:** Verify checkout is inaccessible when not logged in (BR2/FR-12); verify checkout cannot proceed with an empty cart (BR3/FR-13); verify the cart is empty after order confirmation (BR4/FR-14); verify cancel-and-return-to-cart behavior at each step where a Cancel control exists (BR5/FR-15).
- **Boundary testing:** Cart with a single item vs. multiple items; maximum field lengths for First Name/Last Name/Zip (once clarified — see Open Question #5).
- **Cross-browser matrix:** Execute the full flow on Chrome, Firefox, and Safari per NFR-01.
- **Responsive/mobile testing:** Execute the full flow at mobile viewport sizes per NFR-02, once specific breakpoints are clarified.
- **Navigation testing:** Browser back-button behavior at each checkout step, and page-refresh behavior mid-checkout, per Technical Notes.
- **Do not proceed to automated test-case generation for NFR-03 (Usability) and NFR-04 (Security)** until stakeholders provide measurable acceptance criteria — flag these for clarification rather than fabricating pass/fail thresholds.
- **Flag for stakeholder review before test planning:** Open Questions #1, #3, and #4, since they affect whether entire scenarios (payment-method selection, cancel-at-every-step, empty-cart handling) should be in or out of scope for the generated test plan.